How Much Semantic Architecture Is Necessary?

A Causally Controlled A/B/C/D Protocol for Evidence-Constrained Dataset Schema Induction

Anonymous pre-blind technical report
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Abstract

This work studies the semantic component of a system whose product interface is *dataset in, deterministic schema out*. The physical schema remains parser-owned; language models may only propose property-level semantic additions over existing fields and cited evidence. We replace a historically fragmented semantic pipeline with a controlled four-condition experiment: A is deterministic only; B adds at most one dataset-level semantic call; C reuses the exact B response and applies deterministic evidence-identity, field-relevance, and conflict checks; D preserves the legacy per-field/manual-group schedule and non-empty-evidence merge behavior.

The contribution at the present stage is an auditable protocol and validated measurement harness, not a claim that any architecture wins. On nine synthetic development cases, only two exposed dataset-level semantic opportunities and one opportunity was absent from gold. All pairwise architecture comparisons were therefore correctly declared non-comparable. A smaller Qwen3.5 9B backend added no accepted, scorable claim in this set, making the observed B–C difference uninformative rather than a null effect. Adversarial tests established the intended call, replay, evidence, abstention, applicability, unsupported-claim, and AURC invariants.

Backend qualification subsequently exposed a runtime and measurement failure: an 8,192-token Qwen3.6-27B run with thinking effectively enabled was invalidated. After repairing replay and telemetry provenance, disabling thinking, and loading a 16,384-token context, the same Q4_K_M model passed every predeclared operational criterion over eight non-blind calibration datasets and 74 semantic targets. This qualification establishes execution compatibility only. The causal contribution of dataset-level reasoning and deterministic verification remains reserved for an independently annotated, frozen blind evaluation. Explicit provenance placeholders are included for that freeze.

Keywords: semantic schema induction; selective prediction; evidence grounding; deterministic verification; replay identity; model capability debt; dataset metadata.

1 Introduction

Dataset schema induction contains two different epistemic problems. Physical facts such as field names, stored types, shapes, groups, and declared attributes are properties of bytes and format standards. Contextual meaning—whether a column is an identifier, a measurement, a time axis, or a domain quantity—often requires document-level semantic integration. Treating both problems as free-form language generation makes plausible metadata difficult to distinguish from schema truth. Treating both as deterministic parsing leaves genuinely contextual information unused.

The system studied here therefore retains a strict authority boundary. Deterministic extractors own canonical physical structure. A semantic component may propose claims only for existing fields, cite evidence identifiers from the input record, and abstain when support is insufficient. This design is aligned with provenance models that make derivation auditable [7], evidence-backed model revision [1], and selective prediction, where lower coverage can be exchanged for lower risk [3].

The historical implementation used per-field calls, manual grouping, retries, and multiple merge rules. Some of this complexity may have been useful when smaller models struggled to integrate repository- or dataset-level context. We call complexity that remains after its compensating model limitation has weakened *Model Capability Debt*. The scientific question is not whether a newer model can be inserted everywhere. It is whether the minimum architecture that preserves the trust boundary can be identified empirically.

We ask three causal questions:

RQ1: What does one dataset-level semantic reasoning call add beyond deterministic extraction?

RQ2: Holding the semantic response exactly constant, what does deterministic verification change?

RQ3: Does the minimal verified hybrid preserve trustworthiness relative to the legacy per-field system while using fewer semantic calls?

At this pre-blind stage, none of these questions has a valid effect estimate. The completed work establishes that the future estimates can be causally interpreted, and identifies precisely what evidence is still missing.

2 Problem Formulation and Product Boundary

The product-level mapping is

$$\texttt{dataset file(s)} \longrightarrow \texttt{deterministic canonical schema} \oplus \texttt{optional semantic claim ledger}.$$

The deterministic schema is a complete standalone output. Semantic augmentation is an optional overlay and may never invent fields, silently overwrite stronger deterministic claims, or convert missing support into confidence.

For field path f, property p, and proposed value v, a semantic claim is represented as

$$c = (f, p, v, E, s, d),$$

where E is a set of evidence identifiers, s is a confidence value used for risk–coverage ordering, and d is the final decision. Correctness and confidence are deliberately separate: a claim may be confident but unsupported, or supported but uncertain. Decisions include accepted, rejected, and abstained; verification states distinguish verified agreement, evidence-grounded support, contradiction, conflict, and unsupported status.

Gold applicability is likewise explicit. Each field–property slot is one of: (i) applicable with a value, (ii) applicable but legitimately unknown, (iii) not applicable, or (iv) outside an open-world gold scope. This prevents an abstention from being silently scored as an incorrect accepted value and prevents missing gold coverage from being misread as N/A.

3 First-Principles Architecture Audit

The audit separated essential from accidental complexity. Parser authority, structured abstention, property-level provenance, and visible conflict states are essential because they encode trust boundaries. Per-field prompt fragmentation, manual grouping, repeated free-JSON/schema attempts, and redundant transformations are historical until their value is demonstrated. Rigid output contracts remain useful where they make failures observable; they are not justified merely as a way to force a preferred answer.

The audit produced one conservative redesign rule: delete semantic orchestration only after a replay-controlled experiment shows that a simpler approach is equal or better. This avoids replacing old Model Capability Debt with new model dependence.

4 The Four Fixed Architecture Variants

Table 1: Frozen semantic architecture variants. No additional variant is permitted in the study.

ID	Name	Semantics	Semantic generation calls
A	Deterministic only	Emits parser-owned schema and deterministic claims; no semantic model.	0
B	Dataset reasoner	Deterministic extraction plus at most one dataset-level semantic response for all unresolved targets.	≤ 1 per dataset
C	Replayed reasoner + verifier	Consumes the exact B response; checks evidence identity, field relevance, deterministic contradiction, and cross-claim conflict.	0 new calls
D	Legacy semantic pipeline	Preserves per-field/manual-group scheduling, temperature 0.2, free-JSON then schema fallback, and the historical non-empty-evidence merge rule.	Variable, historically many

The B–C contrast is the cleanest intervention. A memoizing backend keys the full request and provides C with the exact raw response and SHA-256 identity produced for B. C is invalid if it generates a response, lacks a replay, or consumes a different hash. Verification is deterministic and does not claim semantic entailment: an existing, field-relevant citation establishes traceable support, not factual truth.

5 Evaluation Methodology

Let V be applicable-value slots, P accepted predictions on those slots, and $K \subseteq P$ correct accepted predictions. We report

$$\text{end-to-end accuracy} = \frac{|K|}{|V|}, \tag{1}$$

$$\text{coverage} = \frac{|P|}{|V|}, \tag{2}$$

$$\text{selective risk} = 1 - \frac{|K|}{|P|}. \tag{3}$$

When P is empty, selective accuracy/risk is undefined rather than forced to zero. Abstentions lower end-to-end accuracy and coverage but do not enter the conditional selective-risk denominator. Confidence is used only to order whole tie groups for a right-continuous risk–coverage curve. AURC is integrated only over achieved coverage; per-property curves are preferred when confidence scales differ.

Unsupported-claim rates are computed over model-proposed or model-accepted claims, never diluted by deterministic claims. Cost accounting separates generated calls, reused upstream calls, and effective architecture cost. A pairwise comparison is non-comparable if either run is partial, replay identity fails, required gold is missing, or a semantic target lies outside the scoring design.

5.1 Adversarial evaluator validation

The focused suites now contain 53 tests across the variant and backend qualification modules, within 332 passing project tests. They cover nonexistent, irrelevant, wrong-span, contradictory, and empty evidence; confidence 0 and 1; malformed and partial output; duplicates and conflicting claims; applicable unknown and N/A; missing or mismatched replay; hidden retries; timeout; zero/all accepted predictions; and confidence ties. Additional regressions protect failure replay identity, per-call context accounting, legacy fallback telemetry, and partial D provenance.

Table 2: Core causal invariants enforced by code and tests.

Invariant	Enforced behavior
A call boundary	Always zero model calls.
B call boundary	At most one dataset-level generation call; hidden retries make the run partial.
C replay boundary	Zero generation calls and exact B raw-response hash, including malformed-response replay.
Evidence boundary	Missing, empty, cross-field, or wrong-span evidence cannot verify a claim.
Confidence boundary	Confidence never implies verification.
Applicability boundary	N/A, applicable unknown, applicable value, and outside-scope are distinct.
Comparability boundary	Partial execution, missing gold coverage, or replay failure suppresses causal deltas.
Legacy fidelity	Historical scheduling and merge behavior remain unchanged; all attempts are measured.

6 Synthetic Development Experiment

The development manifest contains 9 authored regression cases spanning CSV, HDF5, and time-series structures. It is not a random sample and provides no external-validity evidence. Only 2/9 cases contain any dataset-level semantic target. One of those cases targets `/campaign/meta/platform`, which is absent from gold; consequently every formal A–B, B–C, and C–D comparison is non-comparable.

The local development backend was Qwen3.5 9B. The two actual B responses were replayed byte-for-byte into C; the other seven cases correctly made no B/C call. Table 3 reports pooled descriptive diagnostics, not architecture effects.

Table 3: Nine-case synthetic development diagnostics. All B/C/D causal comparisons are non-comparable.

Variant	Comparable	End-to-end acc.	Coverage	Sel. risk	AURC	Calls
A	yes	0.976608	0.988304	0.011834	0.014603	0
B	no	0.976608	0.988304	0.011834	0.014603	2
C	no	0.976608	0.988304	0.011834	0.014603	0 new / 2 effective
D	no	0.988304	1.000000	0.011696	0.014705	6

B proposed one model claim, accepted none, and added no scorable coverage. Its effective cost was 5752 input tokens, 381 output tokens, and 4128.415 ms of model latency. C had the same effective upstream cost and zero new generation. D proposed and accepted four claims, used 6 calls, 7538/3598 input/output tokens, and 22171.453 ms. D recovered two gold-covered logical values in one hard CSV case, but another change was unscored. These facts do not establish superiority because the design is non-comparable.

The Qwen3.5 backend abstained on a postal-code target even though its notes recognized the semantics, and violated target scope on the unscored HDF5 case. A stronger model might resolve either failure without architecture changes; this is a hypothesis, not a result. The development run therefore yields no estimate of the increment from A to B and no verification trade-off from B to C. The all-zero observed B–C decision delta is uninformative, not evidence of no verifier effect.

7 Backend Qualification

Qualification asks whether a backend can execute the frozen contracts, not whether its claims are correct. The fixed non-blind calibration manifest contains 8 external datasets and 74 semantic targets. It is excluded from the future blind benchmark. B is run twice per dataset at temperature 0 and seed 0; C immediately replays each response; D runs once with its preserved policy.

7.1 Invalidated 8K/thinking-on execution

The first Qwen3.6-27B Q4_K_M execution used an 8,192-token loaded context while thinking remained effectively enabled. Only 2/16 B attempts and 1/8 D dataset runs were contract-valid; 12 B attempts terminated at the generation boundary, and a 10,044-token HDF5 prompt was rejected before generation. The run also exposed three measurement defects: malformed B failures did not carry raw identity into C, unparseable output could be counted target-scope valid, and partial D failures dropped earlier-group/fallback provenance. The immutable artifact was invalidated; none of its rates is used for backend selection.

The failure was informative as infrastructure diagnosis. LM Studio 0.4.18 documents a fix for reasoning on/off being ignored by some custom configurations [4]. We repaired only provenance and measurement, not prompts, architectures, targets, or call budgets.

7.2 Corrected 16K/thinking-off execution

A one-call gate first established a contract-valid response with zero reasoning tokens. The same Qwen3.6-27B Q4_K_M model [5] was then run with thinking disabled and a 16384-token context. It passed every frozen criterion (Table 4).

Table 4: Corrected Qwen3.6-27B operational qualification.

Criterion	Observed	Required	Result
Semantic-opportunity datasets	8	≥ 8	pass
B contract-valid rate	1.0	≥ 0.95	pass
B target-scope-valid rate	1.0	1.0	pass
B/C replay identity	1.0	1.0	pass
B exact repeatability	1.0	1.0	pass
B timeout / truncation rates	0.0 / 0.0	0.0 / 0.0	pass
B context fit / observed telemetry	1.0 / 1.0	1.0 / 1.0	pass
D contract-valid rate	1.0	≥ 0.95	pass
D timeout / truncation rates	0.0 / 0.0	0.0 / 0.0	pass
D context fit / observed telemetry	1.0 / 1.0	1.0 / 1.0	pass

All 16 B attempts ended with JSON and `finish_reason=stop`; repeats were byte-identical for all eight datasets. Every C input hash matched B and every C semantic generation count was zero. The maximum observed B context was 13306 tokens; the maximum requested `input+max_tokens` budget was 14142, below the loaded limit.

Table 5: Descriptive operational cost in the corrected qualification.

Condition	Calls	Input tokens	Output tokens	Mean latency
B (two repeats)	16	76976	26436	27.95 s per attempt
C generation	0	0 new	0 new	deterministic replay/verification
D (one run)	59	79571	19547	44.61 s per dataset

D’s 59 calls are an observed consequence of historical scheduling: the HDF5 case alone required 22 field calls. One `functional_traits.csv` free-JSON shape attempt failed and the preserved schema fallback succeeded. The cost gap is real; an accuracy–cost trade-off has not yet been estimated.

8 What the Current Results Do and Do Not Establish

Established facts. The harness enforces the four fixed architectures, exact B/C replay, zero C generation, fail-closed contracts, explicit applicability, and per-call telemetry. The Qwen3.6-27B Q4_K_M condition is operationally eligible under the corrected runtime. The legacy architecture has a substantially larger call surface on the calibration tasks.

Not established. The study has no blind effect estimate. The nine-case development set is too sparse and incompletely gold-covered to estimate A–B, B–C, or C–D effects. Backend qualification contains no gold. It cannot show that dataset-level reasoning helps, that verification lowers risk, that D is more accurate, or that Qwen generalizes.

Model selection. There is no operational reason to replace Qwen3.6-27B with GLM, Llama, or a smaller Qwen. Alternative families may be predeclared sensitivity conditions, but adding them post hoc to rescue an unfavorable architecture result would confound backend and architecture selection. Model cards and exact runtime identity should be reported as first-class artifacts [6].

9 Blind External Evaluation Protocol

Two annotators who did not develop the system will independently label the blind gold. They will distinguish applicable value, applicable unknown, N/A, and outside-scope slots, then adjudicate disagreements without observing architecture outputs. Dataset-level paired effects, not pooled slots alone, will be the primary analysis unit. The development set will not be used as evidence of generalization. Dataset documentation follows the broader principle that collection, intended use, and limitations should be explicit [2].

The primary contrasts are predeclared:

- A–B: added correct/incorrect claims, coverage, selective risk, unsupported claims, calls, tokens, and latency;
- B–C: B-accepted claims rejected or abstained by C, incorrect claims removed, correct claims lost, unsupported claims removed, and the risk–coverage trade-off;
- C–D: quality, selective metrics, unsupported claims, calls, tokens, latency, and failure surface, only when D is comparable.

9.1 Blind-freeze provenance placeholders

STOP: the fields below are intentionally unresolved.
They must be content-addressed and countersigned before blind gold is unsealed. Leaving a field blank blocks interpretation; it must never be silently replaced by the development or qualification artifact.

Blind sampling registration receipt	<TO BE BOUND BEFORE GOLD UNSEALING>
Blind candidate-frame SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Blind manifest SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Independent annotation package SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Adjudicated gold package SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Annotation vocabulary/handbook SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Annotator pseudonyms and independence attestations	<TO BE BOUND BEFORE GOLD UNSEALING>
Calibration-gate result and agreement statistics	<TO BE BOUND BEFORE GOLD UNSEALING>
Frozen backend-registry SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Exact Qwen GGUF checkpoint SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
LM Studio 0.4.18 runtime/version receipt SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Frozen source commit/tree SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Frozen analysis-plan SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Prompt/schema bundle SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Gold unsealing timestamp and countersignatures	<TO BE BOUND BEFORE GOLD UNSEALING>

The known pre-freeze contract hashes are reported in Appendix A; they do not substitute for the final registry, checkpoint, source-tree, or gold receipts.

10 Threats to Validity

Construct validity. Evidence identity and field relevance do not prove semantic entailment. C should be interpreted as deterministic trust-boundary verification, not an oracle. Confidence values may be incomparable across properties; per-property selective curves are therefore necessary.

Internal validity. B/C replay is tightly controlled, but D intentionally retains temperature 0.2 and historical fallback behavior. Backend state, runtime version, context, checkpoint, and prompt hashes must remain frozen. Any partial response or missing replay suppresses the corresponding comparison.

External validity. The nine development cases are synthetic and the eight qualification cases are a fixed operational calibration set already exposed to development. Neither is a population sample. Only the future independently annotated corpus can support generalization claims.

Statistical conclusion validity. The blind analysis must use paired dataset-level effects and confidence intervals. A clean null result is acceptable. The current manuscript reports descriptive pooled metrics only to document harness behavior.

Reproducibility. The exact checkpoint-file hash and machine-observed LM Studio version are still missing. The version is operator-reported. These are explicit freeze blockers rather than hidden caveats.

11 Conclusion

The durable system objective remains dataset input to deterministic schema output, with optional evidence-constrained semantic enrichment. The present work reduces the semantic architecture question to four auditable variants and validates the measurement boundaries needed for causal interpretation. Development data do not rank the variants. Corrected backend qualification shows that one dataset-level Qwen3.6-27B response can be generated repeatably and replayed into deterministic verification, while the legacy condition exposes a much larger call surface.

The central research answer is intentionally deferred: whether semantic reasoning adds useful claims, and whether deterministic verification removes more errors than correct claims, can only be learned after the blind provenance block is completed and independent gold is unsealed. Until then, the scientifically correct result is a validated protocol, an operationally eligible backend, and no architecture winner.

A Content-Addressed Current Artifacts

Artifact	SHA-256
Development manifest	`a1a199272b9019dd99dab873239f18ba1d8b656235c54ac40ad84fb5d1738797`
Development report	`1e77f66d924622153798743c209497b5025271c5d94ad4cebf6290871ff87e27`
Qualification manifest	`71aa69e25fddf8a3cfe9ad78e2cc2ba08015347f8ea53166985c081120fe2666`
Controlled vocabulary	`874548a10f8871449be9b749c99e95e72e8852d261324abc8fa3c492246ed888`
Invalidated 8K qualification	`d0a5fb9a82d41bda557669fd384820485c5265f187679f2e954ad347eec25cde`
Corrected qualification	`c89145cf23943fd2991504a31cea5a10ceddccc90725e31932cc02770b1c0538`
Qualification assessment	`bec8279af416b796d4ea5fe071a8947fcd1d7d123c1cc021168dd928fcb87efb`
Dataset reasoner prompt	`1b75583a35549a5cbdfecc79be5796eb358d2e5f976f5483b25756a81f8f8060`
Dataset response schema	`e79212874554daf93c401578b97963f45934854b22b6786ee41ae60bf8dfd7ca`
Legacy prompt	`3d4047f7bdc36e6ae832fe7e988bd06bff39be81f73bea101c6abdfe32b78a0f`
Legacy response schema	`83ee4fb51579af04a304c61eb4b610263b1d02fcae1a90a56cfdb131e445a91f`

B Current Reproduction Status

At manuscript generation, 332 project tests passed. The relevant Ruff checks passed, all 316 JSON artifacts parsed successfully, and `git diff --check` passed. Seven pre-existing full-tree lint findings in unrelated legacy files remain documented and were not modified. The two PDFs were rendered from these TeX sources using Tectonic 0.16.9 [8].

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